

Methodology: Pipeline Design and Formal Definitions

This section describes the five-stage computational meta-analysis pipeline. Each stage corresponds to a tested, independently executable script that reads upstream outputs and produces structured artifacts. The pipeline extends the systematic literature analysis approach of Knight et al. knight2022sep—which combined manual annotation with ontology-based automated analysis—by substituting manual coding with fully automated, LLM-driven assertion extraction and citation-weighted hypothesis scoring.

Pipeline Overview

Stage	Script	Primary Input	Primary Output	Section
1	01_literature_api_search.py	API queries	corpus.json	Retrieval
2	02_meta_analysis.py	corpus.json	Classification, temporal, TF-IDF, NMF	Bibliometrics